PHYS 209: Aspects of Modern Physics

# Module 4: Qubits

## Lecture 1: Mathematics of Qubits

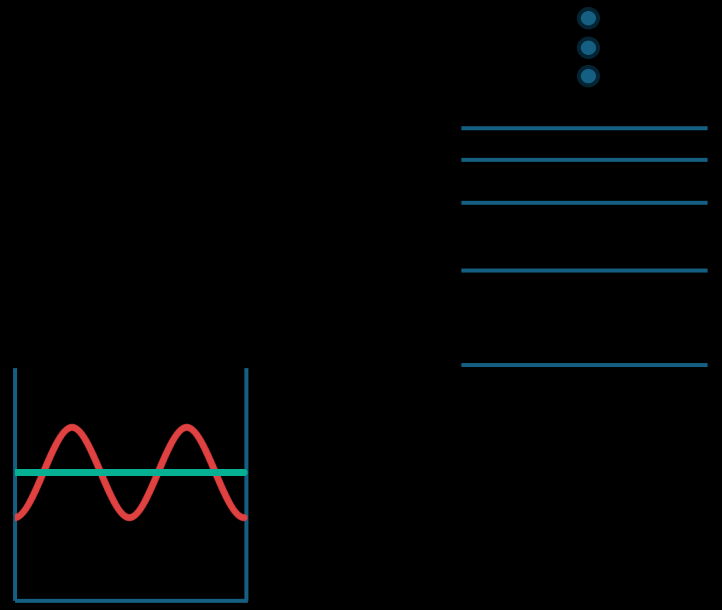
Instructor: Muhammad Hani Zaheer

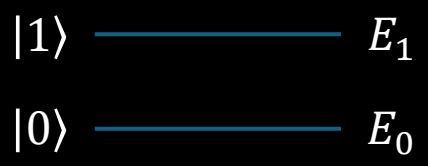
# Outline

- This is the 2nd quantum module covering two level systems / qubits.
- General (tentative) module outline:
  - Mathematics of qubits
  - Quantum sensing with qubits
  - Quantum entanglement and local realism
  - No cloning theorem and quantum teleportation
  - Quantum computation with qubits

# Review: Quantum Mechanics

- Quantum systems have quantized energies.
- Quantum states are described by kets $|\psi\rangle$.
  - For infinite dimensional systems, $|\psi\rangle$ can be represented by a wave function.
    - Ex: Particles in a box.
  - For finite dimensional systems, $|\psi\rangle$ can be represented by a column matrix.
    - Ex: Two Level System / Qubit.





# Two Level Systems: Matrix Representation

- Two Level Systems: two possible energy states
- Matrix Representation (kets):

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- Bra: Complex conjugate + transpose of kets

$$\langle 0| = [1 \quad 0], \qquad \langle 1| = [0 \quad 1]$$

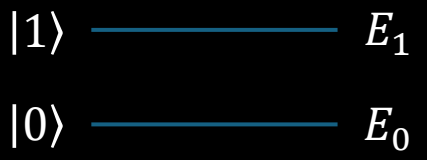
- Inner product:

$$\langle 0|0\rangle = [1 \quad 0]\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 + 0 = 1$$

$$\langle 0|0\rangle = \langle 1|1\rangle = 1 \qquad \text{Normalized}$$

$$\langle 0|1\rangle = \langle 1|0\rangle = 0 \qquad \text{Orthogonal}$$

  - These states form an orthonormal basis set.

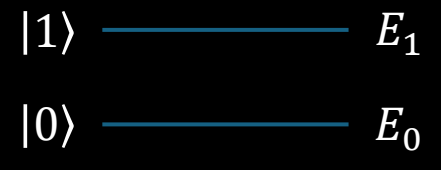


# General Qubit State

- Qubits can be in a superposition state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha\begin{bmatrix}1\\0\end{bmatrix} + \beta\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}\alpha\\\beta\end{bmatrix}$$

$$\langle\psi| = [\alpha^* \quad \beta^*]$$

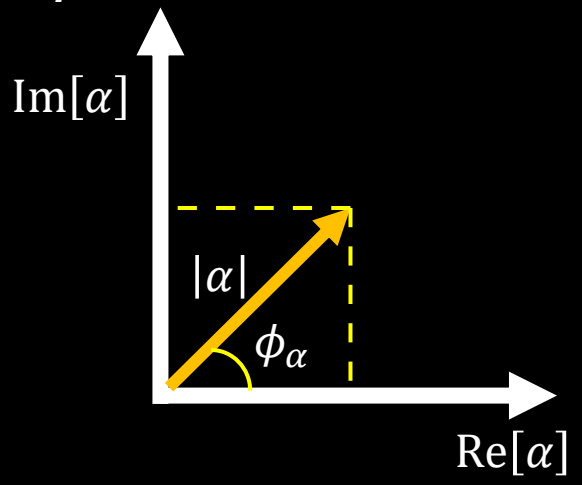



- Here $\alpha$ and $\beta$ can be complex numbers:

$$\alpha = \mathrm{Re}[\alpha] + i\,\mathrm{Im}[\alpha], \qquad \beta = \mathrm{Re}[\beta] + i\,\mathrm{Im}[\beta]$$

- Equivalently, they can be written in polar form

$$\alpha = |\alpha|e^{i\phi_\alpha}, \qquad \beta = |\beta|e^{i\phi_\beta}$$



# Qubit Matrix Representation

- Matrix Representation:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \qquad \langle\psi| = [\alpha^* \quad \beta^*]$$

  - Bra = Complex Conjugate + Transpose of Kets

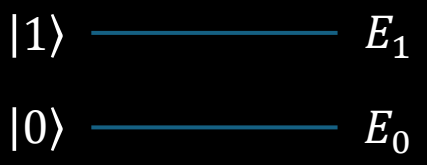
- Normalization:

$$\langle\psi|\psi\rangle = 1$$

$$[\alpha^* \quad \beta^*] \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = 1$$

$$\alpha^*\alpha + \beta^*\beta = 1$$

Sum of Probabilities = 1




Remember:

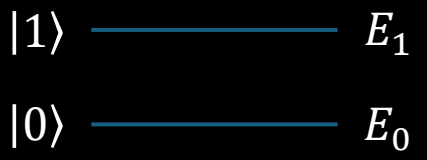
$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx = 1$$

# Qubit Probabilities

- Matrix Representation:

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}, \qquad |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

$$|\psi\rangle = \begin{bmatrix}\alpha\\\beta\end{bmatrix}, \qquad \langle\psi| = [\alpha^* \quad \beta^*]$$




- Probabilities

$$P(0) = |\langle 0|\psi\rangle|^2 = \left|[1 \quad 0]\begin{bmatrix}\alpha\\\beta\end{bmatrix}\right|^2 = |\alpha|^2 = \alpha^*\alpha$$

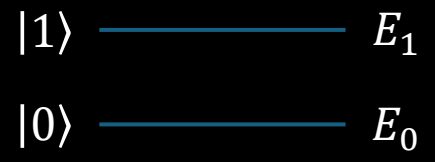
$$P(1) = |\langle 1|\psi\rangle|^2 = \left|[0 \quad 1]\begin{bmatrix}\alpha\\\beta\end{bmatrix}\right|^2 = |\beta|^2 = \beta^*\beta$$

Remember (normalization):
$\alpha^*\alpha + \beta^*\beta = 1$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix}\alpha\\\beta\end{bmatrix}$$

# How many real parameters does a qubit have?

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix}\alpha \\ \beta\end{bmatrix}$$




- 2 complex parameters: $\alpha, \beta$
- 4 real parameters: $\mathrm{Re}[\alpha], \mathrm{Im}[\alpha], \mathrm{Re}[\beta], \mathrm{Im}[\beta]$
- Constraints:
  - Normalization:

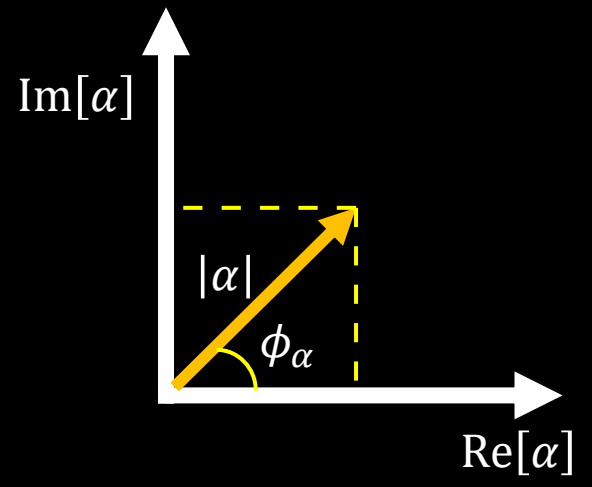
$$\alpha^*\alpha + \beta^*\beta = 1$$

  - Global Phase is unobservable: The following is identical to above.

$$|\psi\rangle = |\alpha||0\rangle + |\beta|e^{i(\phi_\beta - \phi_\alpha)}|1\rangle$$

- Result: Only 2 real parameters

# Bloch Sphere Representation




- Use polar representation:

$$\alpha = |\alpha|e^{i\phi_\alpha}, \qquad \beta = |\beta|e^{i\phi_\beta}$$

- Use global phase freedom to set $\phi_\alpha = 0$ and $\phi_\beta = \phi$.

$$|\psi\rangle = |\alpha||0\rangle + |\beta|e^{i\phi}|1\rangle$$

- The state can be written as

$$|\psi\rangle = \begin{bmatrix} |\alpha| \\ |\beta|e^{i\phi} \end{bmatrix}, \qquad \langle\psi| = [|\alpha| \quad |\beta|e^{-i\phi}]$$

- Normalization:

$$1 = \langle\psi|\psi\rangle = [|\alpha| \quad |\beta|e^{-i\phi}] \begin{bmatrix} |\alpha| \\ |\beta|e^{i\phi} \end{bmatrix} = |\alpha|^2 + |\beta|^2$$

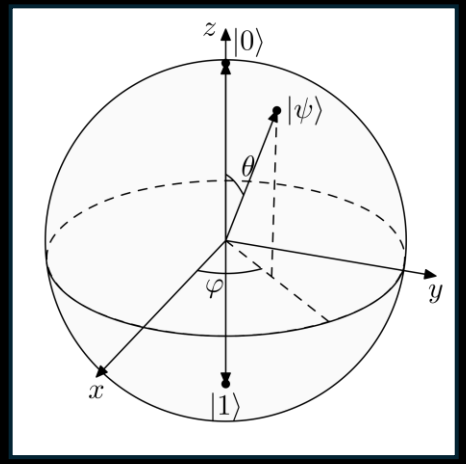
- We can thus parameterize $|\alpha|, |\beta|$ with a single parameter. We choose

$$|\alpha| = \cos\frac{\theta}{2}, \qquad |\beta| = \sin\frac{\theta}{2}$$

# Bloch Sphere Representation

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi}|1\rangle$$

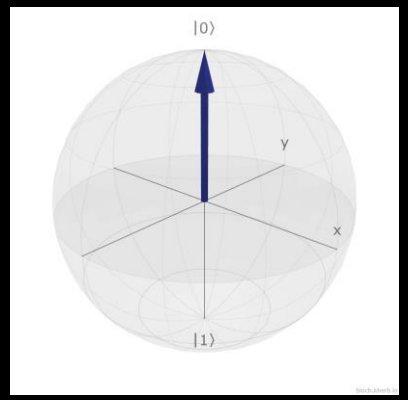
- Here $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$
  - These are exactly equivalent to polar angle and azimuthal angle of a unit sphere.
  - In the context of qubits, this is called the Bloch sphere.



# Bloch Sphere Representation

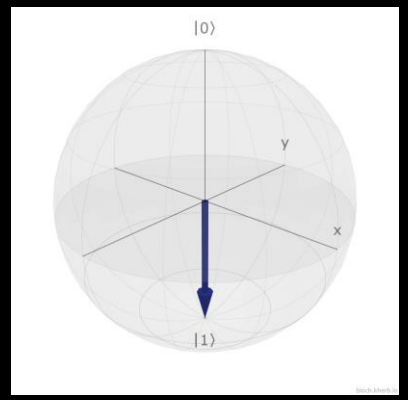
$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi}|1\rangle$$

$\theta = 0$




$|0\rangle$

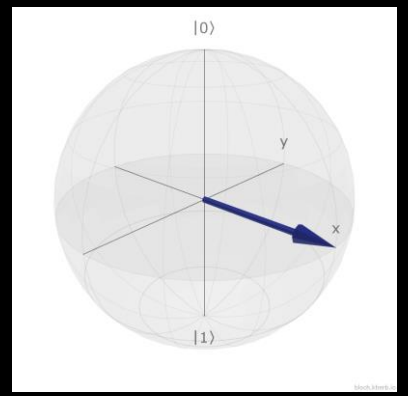
$\theta = \pi$




$|1\rangle$

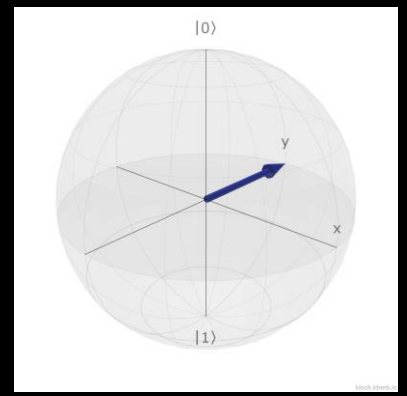
The North and South are the basis states.

$\theta = \frac{\pi}{2}, \phi = 0$




$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

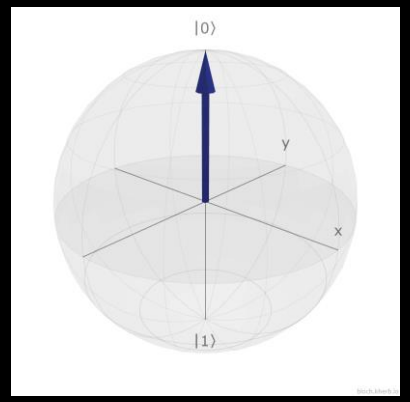
$\theta = \frac{\pi}{2}, \phi = \frac{\pi}{2}$



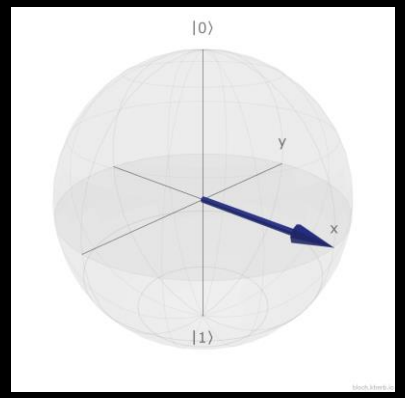

$$\frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

The equator represents states of maximum superposition.

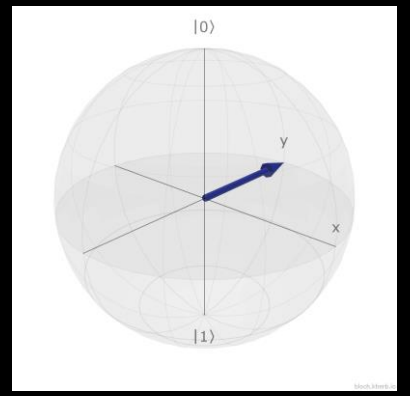
# Antipodal Points are Orthogonal



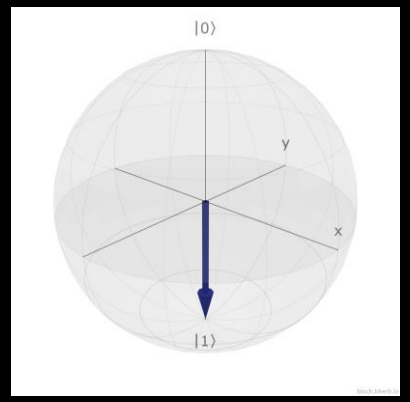

$|0\rangle$



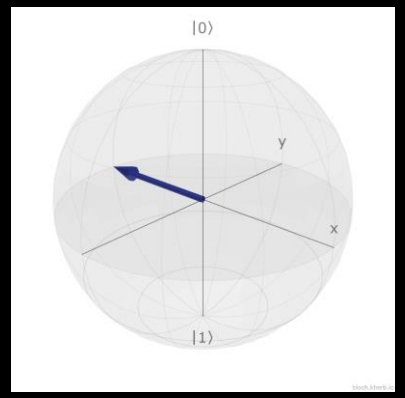

$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$



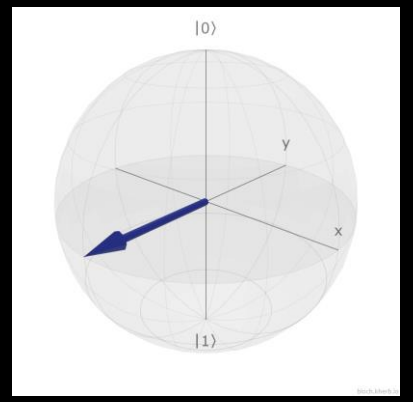

$\frac{|0\rangle + i|1\rangle}{\sqrt{2}}$




$|1\rangle$




$\frac{|0\rangle - |1\rangle}{\sqrt{2}}$




$\frac{|0\rangle - i|1\rangle}{\sqrt{2}}$

# Pauli Matrices

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Consider a general state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

- The NOT gate $X$ switches the bits

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 0 \\ 1 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$X|\psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

# Pauli Matrices

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Consider a general state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

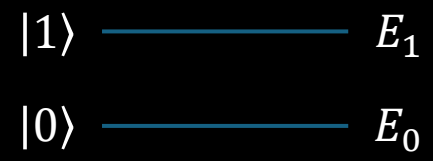
- The phase flip gate $Z$ flips the phase

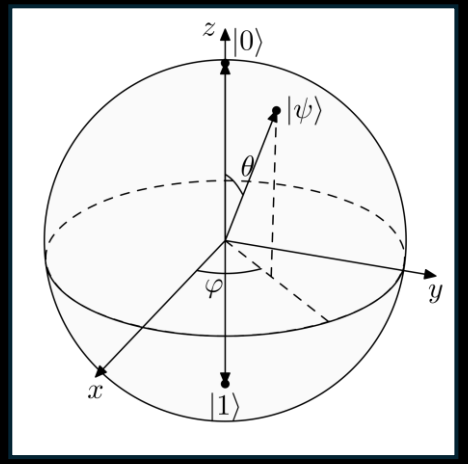
$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 \\ 0 \cdot 1 + (-1) \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0 + 1 \cdot 1 \\ 0 \cdot 0 + (-1) \cdot 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -|1\rangle$$

$$Z|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

Go to: bloch.kherb.io/




# Thank you for your attention!